Activity: thinking about data

In this class activity, you will continue thinking about the data from homework 6.

Important: The purpose of the research study described here (and the associated assignments and class activities) is not to claim that some group compositions inherently work better than others – that would be a prescriptivist approach. Rather, the researchers are interested in *describing* how people interact. This is a purely observational study designed to understand what interactions and group dynamics look like in practice.

Recall that researchers collected data on 74 groups of MBA students at Columbia University. Each group had 3–6 members, and was tasked with completing a week-long project on maximizing financial performance of a fictional blood-testing laboratory. The data here represents a subset of the data made available by the researchers. Each row in the data represents one group, with the following information:

- *performance*: the final performance of the group in the project
- *size*: the number of group members (between 3 and 6)
- *females*: the number of female group members
- *diversity*: a measure of group diversity that summarizes information on the gender, ethnicity, and country of origin for the group members. Takes on values between 0 and 1 (in this data, the values range from 0.25 to 0.65). Higher diversity scores mean the group is more diverse.
- *testosterone*: the average testosterone levels (in pg/mL) for the group members

Data collection

While the data reported here are at the group level, the researchers first had to collect data on each student, before aggregating for each group. The student-level data contains information on 370 participants total, and records their age, gender, ethnicity, country of origin, and testosterone levels.

- Testosterone levels were collected one week before the group task, using a saliva sample from each participant, and were measured in pg/mL
- For gender, only “male” and “female” appear in the dataset; “other” was an option but not selected by any respondents

- For ethnicity, the observed responses in the data include “Asian,” “Black,” “Hispanic,” “Other,” “South Asian,” “South East Asian,” and “White”; it is not clear if other options were available to respondents

Discussion questions

Consider each of the following discussion questions with your neighbor. You do not need to write anything down here, just talk through your thoughts.

1. Why do you think I only shared the group-level data with you, rather than the student-level data?
2. Several of the group-level variables involve aggregating information across students in a group. For example, averaging testosterone values, and counting the number of female students in the group. Group diversity, however, seems harder to measure. How might you construct a measure for diversity, using the available information?
3. Below are demographic characteristics for two of the groups in the data. Which group appears more homogeneous?

Team ID	Age	Gender	Ethnicity	Country
5	29	Female	White	USA
	27	Male	White	Netherlands
	28	Male	Hispanic	Mexico
	24	Female	South Asian	India
65	26	Male	Asian	Korea
	27	Male	White	USA
	27	Female	White	USA
	26	Female	White	USA